EXPERIMENTS

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COURSE CODE:**DSA0601 DATA HANDLING ANDVISUALIZATION**

SET 3 – PRODUCT SALES ANALYSIS

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

QUESTION 1

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

AIM

To create a grouped bar chart to visualize the January, February, and March sales of each product.

PROCEDURE

1. Create the monthly product sales data.
2. Store the sales of each product for January, February, and March.
3. Arrange the data into a matrix.
4. Use the barplot() function.
5. Set beside = TRUE to create grouped bars.
6. Add axis labels, title, and legend.

CODE

```
product <- c("Product A", "Product B", "Product C")

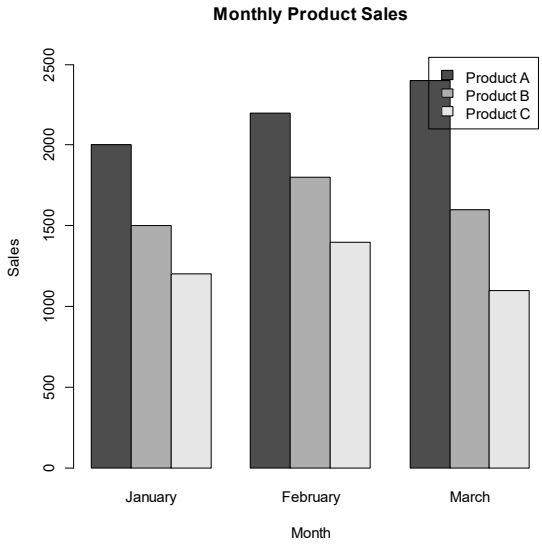
sales <- matrix(
  c(2000, 2200, 2400,
    1500, 1800, 1600,
    1200, 1400, 1100),
  nrow = 3,
```

```
  byrow = TRUE
)

colnames(sales) <- c("January", "February", "March")
rownames(sales) <- product

barplot(
  sales,
  beside = TRUE,
  names.arg = c("January", "February", "March"),
  xlab = "Month",
  ylab = "Sales",
  main = "Monthly Product Sales",
  legend.text = rownames(sales),
  ylim = c(0, 2600)
)
```

OUTPUT



RESULT

Thus, the grouped bar chart was successfully created to visualize the first-quarter sales of each product.

QUESTION 2

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

AIM

To create a stacked area chart showing the sales trend of all products over January, February, and March.

PROCEDURE

1. Create the monthly sales data.
2. Store the sales of each product.
3. Calculate cumulative sales for stacking.
4. Create the area chart using polygon().
5. Add labels, title, and legend.

CODE

```
months <- c("January", "February", "March")

product_a <- c(2000, 2200, 2400)
product_b <- c(1500, 1800, 1600)
product_c <- c(1200, 1400, 1100)

total_ab <- product_a + product_b
total_abc <- product_a + product_b + product_c

x <- 1:3

plot(
  x,
  total_abc,
  type = "n",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales",
  main = "Stacked Area Chart of Product Sales",
  ylim = c(0, 6000)
)

axis(
  1,
  at = x,
  labels = months
)

polygon(
```

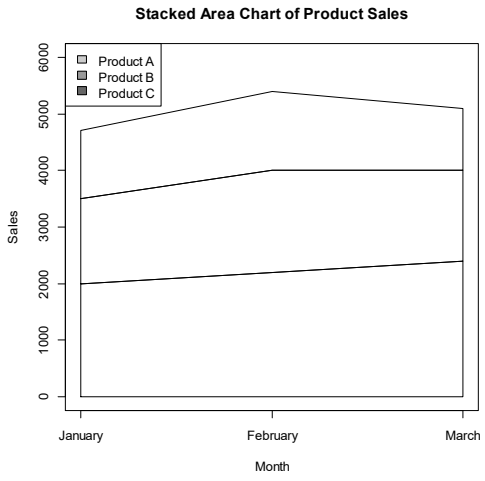
```
  c(x, rev(x)),
  c(rep(0, 3), rev(product_a)),
  border = "black"
)

polygon(
  c(x, rev(x)),
  c(product_a, rev(total_ab)),
  border = "black"
)

polygon(
  c(x, rev(x)),
  c(total_ab, rev(total_abc)),
  border = "black"
)

legend(
  "topleft",
  legend = c("Product A", "Product B", "Product C"),
  fill = c("grey80", "grey60", "grey40")
)
```

OUTPUT



RESULT

Thus, the stacked area chart was successfully created to represent the overall sales trend of all products over the three months.

QUESTION 3

Build a table to show the monthly sales data for each product. Label the table elements.

AIM

To create a table displaying the monthly sales of each product.

PROCEDURE

1. Create the product sales data.
2. Store Product ID, Product Name, January, February, and March sales.
3. Create a data frame.
4. Display the data frame as a table.

CODE

```
product_id <- c(1, 2, 3)

product_name <- c(
  "Product A",
  "Product B",
  "Product C"
)

january <- c(2000, 1500, 1200)

february <- c(2200, 1800, 1400)

march <- c(2400, 1600, 1100)

sales <- data.frame(
  Product_ID = product_id,
  Product_Name = product_name,
  January_Sales = january,
  February_Sales = february,
  March_Sales = march
)

print(sales)
```

OUTPUT

```
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1    Product A          2000           2200        2400
2          2    Product B          1500           1800        1600
3          3    Product C          1200           1400        1100
```

RESULT

Thus, a stacked area chart was successfully created to visualize the distribution of website user interactions.

QUESTION 4

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

AIM

To develop an interactive product sales dashboard using R Shiny containing a grouped bar chart, stacked area chart, and sales table.

PROCEDURE

1. Install the shiny and ggplot2 packages.
2. Create the monthly product sales dataset.
3. Create a user interface with a product filter.
4. Create a grouped bar chart.
5. Create a stacked area chart.
6. Create a sales table.
7. Display all three outputs in the dashboard.
8. Run the Shiny application.
9. Use the product filter to interact with the data.

CODE

```
library(shiny)
library(ggplot2)

sales <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

long_data <- data.frame(
  Product = rep(sales$Product_Name, each = 3),
  Month = rep(c("January", "February", "March"), 3),
  Sales = c(
    sales$January,
    sales$February,
    sales$March
  )
)

ui <- fluidPage(
  titlePanel("Product Sales Analysis Dashboard"),
```

```
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "product",
        "Select Product:",
        choices = c("All", sales$Product_Name),
        selected = "All"
      )
    ),

    mainPanel(
      plotOutput("barChart"),
      plotOutput("areaChart"),
      tableOutput("salesTable")
    )
  )
)

server <- function(input, output) {

  filtered_data <- reactive({
    if (input$product == "All") {
      long_data
    } else {
      long_data[long_data$Product == input$product, ]
    }
  })

  output$barChart <- renderPlot({
    ggplot(
      filtered_data(),
      aes(x = Month, y = Sales, fill = Product)
    ) +
      geom_bar(
        stat = "identity",
        position = "dodge"
      ) +
      labs(
        title = "Grouped Bar Chart of Product Sales",
        x = "Month",
        y = "Sales"
      )
  })

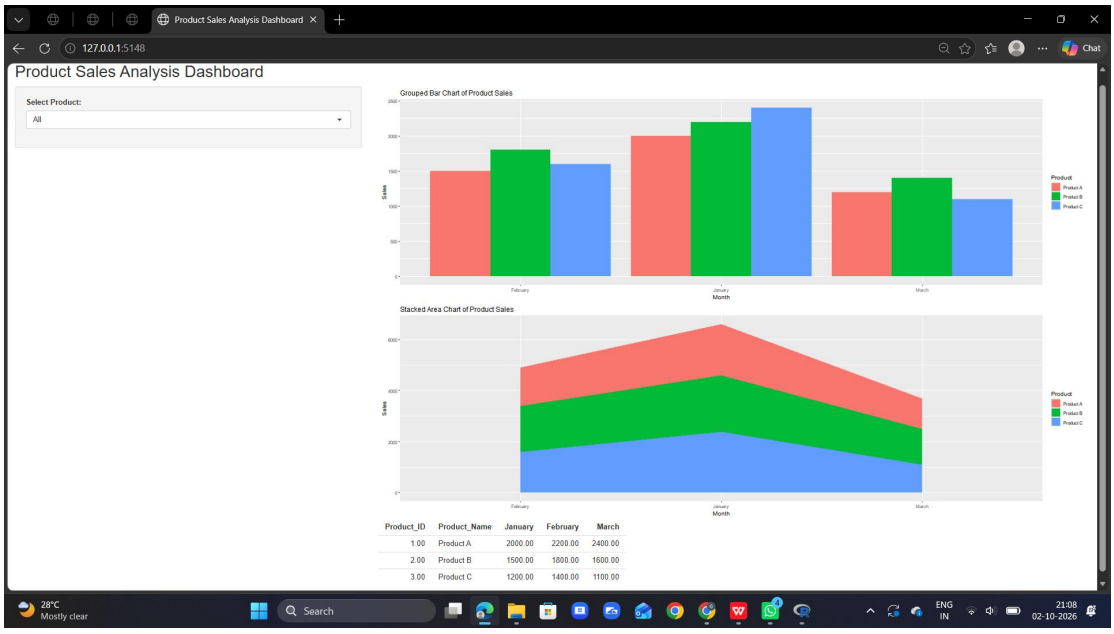
  output$areaChart <- renderPlot({
    ggplot(
      filtered_data(),
      aes(x = Month, y = Sales, fill = Product, group = Product)
    ) +
      geom_area(position = "stack") +
```

```
    labs(
      title = "Stacked Area Chart of Product Sales",
      x = "Month",
      y = "Sales"
    )
  })

  output$salesTable <- renderTable({
    if (input$product == "All") {
      sales
    } else {
      sales[sales$Product_Name == input$product, ]
    }
  })
}

shinyApp(ui = ui, server = server)
```

OUTPUT



RESULT

Thus, an interactive **Product Sales Analysis Dashboard** was successfully developed using **R Shiny**, containing a grouped bar chart, stacked area chart, and monthly sales table.